3-1. A first-order discrete system is described by the following equation:

$$\mathrm{x(k+1)} = -0.5\mathrm{x(k)} + \mathrm{u(k)}.$$

The performance measure to be minimized is

$$\mathrm{J} = \sum_{k=0}^{2} |x(k)|,$$

and the admissible states and controls are constrained by

$$-2 \le \mathrm{x(k)} \le 2, \mathrm{k} = 0,1,2,$$
$$-1 \le \mathrm{u(k)} \le 1, \mathrm{k} = 0,1.$$

a) Carry out by hand the computational steps required to determine the optimal control law by using dynamic programming. Quantize u(k) and x(k) in steps 0.5 and 1, respectively, about zero, and use linear interpolation.

b) What is the optimal control sequence for an initial state value of 1?

3-2. Give the detailed derivation of the recurrence relation of the dynamic programming, i.e. the Bellman' s recurrence equation.

3-3. It was mentioned on 84 of the reference book written by Kirk that ``F(N-k) goes to infinity as k goes to infinity". However, in example 3.10-1 on Page 84, the plots for the optimal control gains are subject to sudden increases, see figure 3-9(a).

(1) Give the optimal control sequence for k=0, 1, 2, 3, and 4 with the minimum cost values.

(2) Please explain why the control gains suddenly increases after t=150s in the figure, with Matlab codes, the resulted data and figure plots for k>=300.

3-4. Please give the solution to excises 3.8 on page 99, with a detailed derivation procedure.

3-5. Please give the solution to excise 3.10 on page 99, with a detailed derivation procedure.

3-6. The first step in using the Hamilton-Jacobi-Bellman equation

$$0 = J_t^*(x(t), t) + \min_{u(t)} \{g(x(t), u(t), t) + J_x^*(x(t), t)[a(x(t), u(t), t)]\}$$

is to determine the admissible control $u^*(t)$ [in terms of x(t), t and J_x^*] that minimizes $\{\cdot\}$. Find $u^*(t)$—expressed as a function of x(t), t and J_x^*, for the system

$$\dot{x}_1(t) = 0.5x_2(t),$$
$$\dot{x}_2(t) = -0.5x_1(t) + x_2(t) + u(t),$$

and the performance measure

$$\mathrm{J} = \int_0^T \frac{1}{2}\left[q_1 x_1^2 + q_2 x_2^2 + u^2(t)\right]dt,\ q_1, q_2 > 0.$$

The admissible controls are constrained by

$$|\mathrm{u(t)}| \le 1.$$